



국립한국교통대학교

LECTURE #2

LONGITUDINAL DYNAMICS

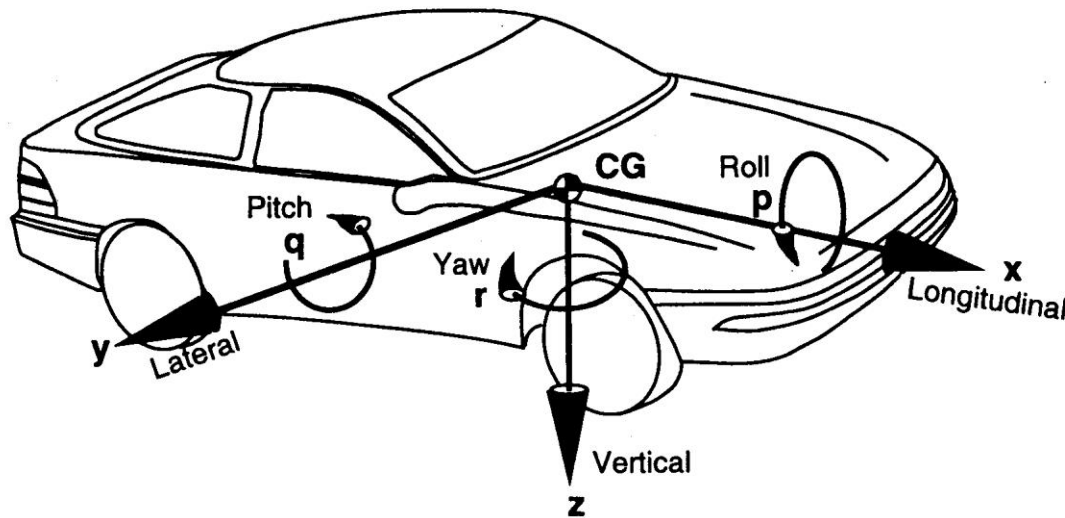
Automotive Engineering

KNUT

Spring, 2024

Vehicle Fixed Coordinate System

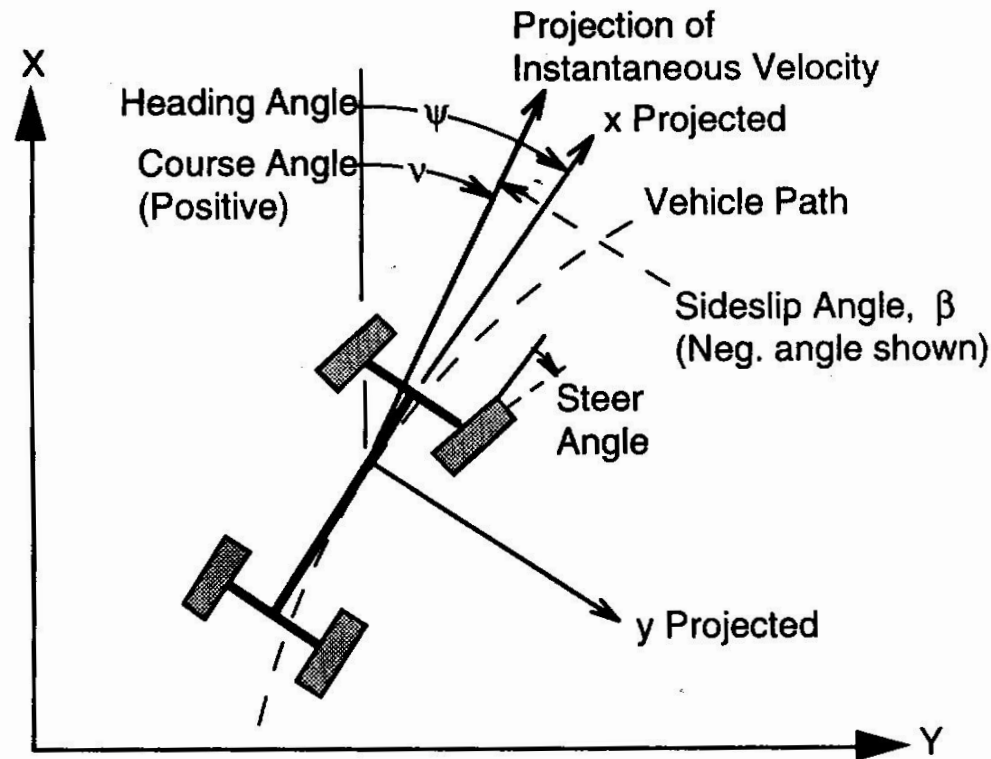
- The vehicle motions are defined with reference to a right-hand orthogonal coordinate system



SAE Vehicle Axis System

Earth Fixed Coordinate System

- Vehicle attitude and trajectory are defined with reference to a right-hand orthogonal coordinate system fixed on the earth



Vehicle in an earth fixed coordinate system

▶ Vehicle Chassis System

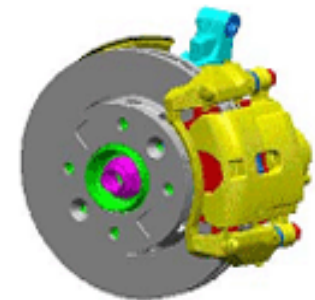
- Definition: **Frame** plus **running gears** like engine, transmission, drive shaft, differential gear and suspension **excluding body**

Chassis system

Suspension system
현가장치

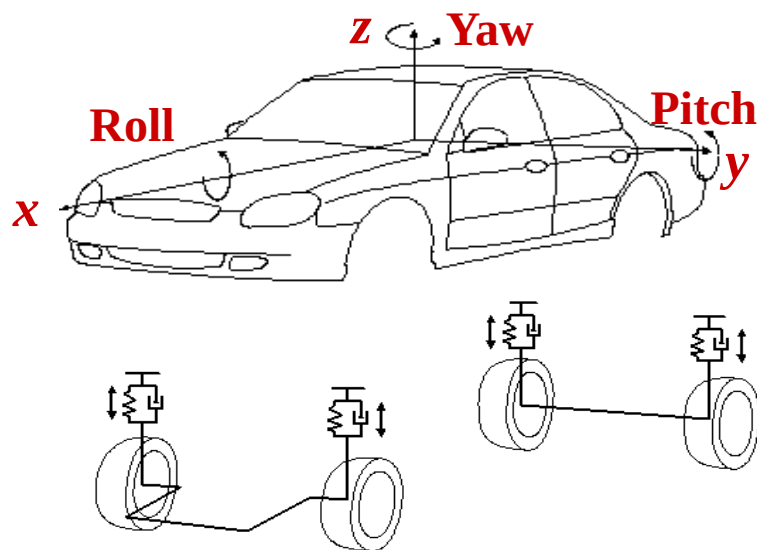
Steering system
조향장치

Brake system
제동장치

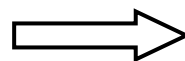


▶ Vehicle Dynamics: Modeling

- Degree of freedom (DOF) ~ number of states
 - 하나의 완성된 장치에 또는 기구에 포함된 모든 계들을 완전히 정지시키기 위해 필요한 제어 입력의 개수
 - 시스템의 동역학 모델링(미분방정식)에 포함된 독립변수의 개수



- Equation of motion:
 - Free body diagram(자유물체도)
 - Force balance
 - Moment balance



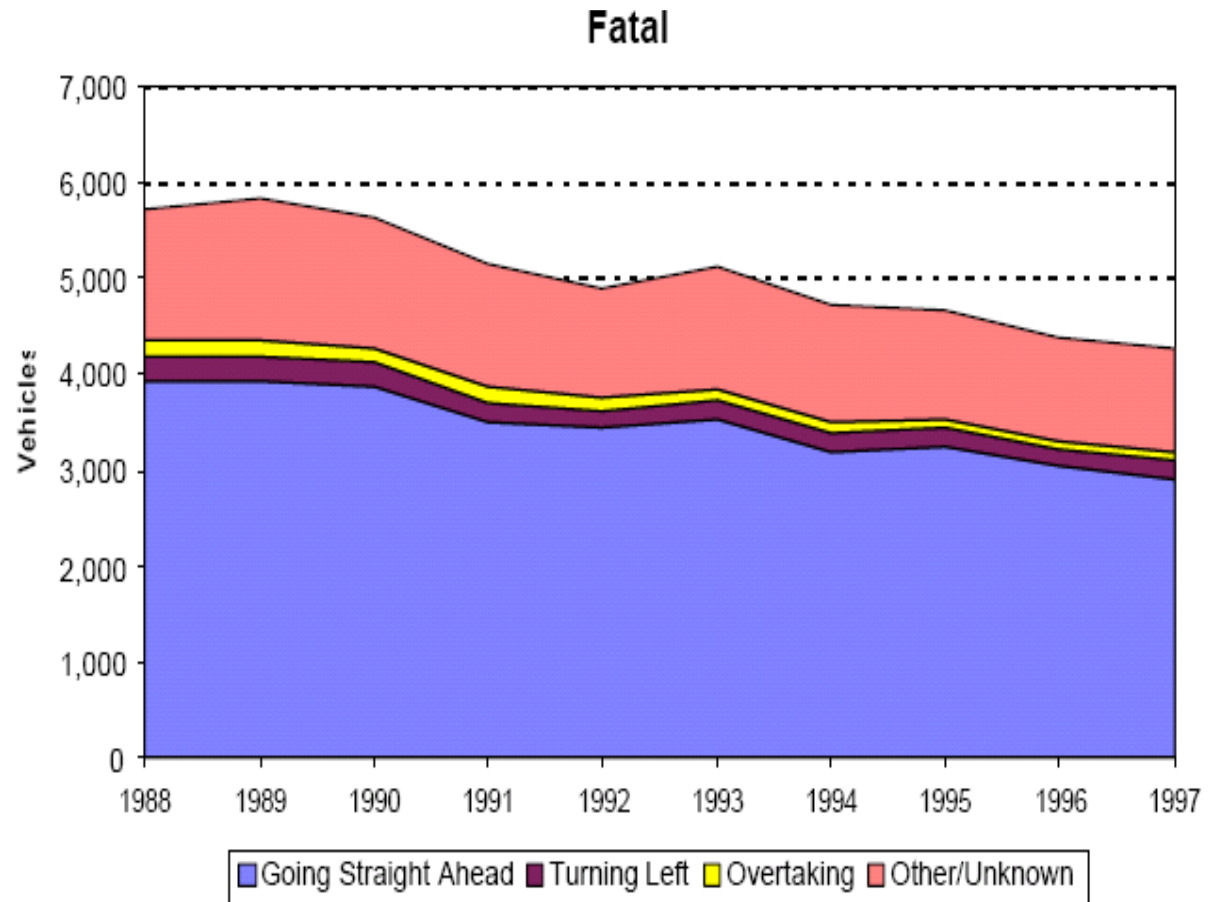
Differential equation

Longitudinal Dynamics

- Longitudinal dynamics matters: main cause of accidents

Vehicle Manoeuvre	1997
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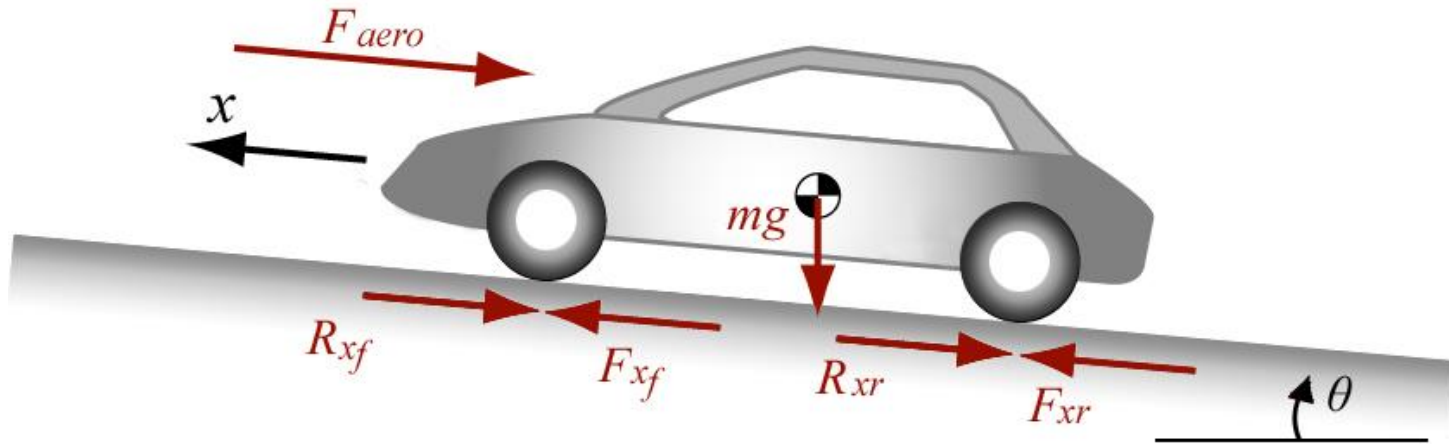
Going Straight Ahead	141,511
Turning Left	25,013
Turning Right	7,168
Making U-turn	811
Changing Lanes	3,489
Merging	1,687
Reversing	2,386
Overtaking	2,340
Slowing/Stopping	16,546
Stopped in Traffic	26,560
Starting in Traffic	2,335
Starting from Parked	537
Entering Parked Position	327
Parked Legally	3,940
Parked Illegally	161
Swerving to Avoid Object	363
Other/Unknown	39,247
Total Vehicles	274,421



http://www.tc.gc.ca/roadsafety/tp/tp13743/2000/pdf/Trends_88-97.pdf

► Longitudinal Dynamics

- Equations of motion: Force balance



$$m\ddot{x} = -F_{aero} + F_{xf} + F_{xr} - (R_{xf} + R_{xr}) - mg \sin\theta$$

F_{aero} : equivalent longitudinal drag force

$F_{xf/xr}$: longitudinal tire force at the front/rear tires

$R_{xf/xr}$: rolling resistance at the front/rear tires

θ : road inclination angle

► Aerodynamics

- Aerodynamic drag force

$$F_{\text{aero}} = -\frac{1}{2}\rho C_d A_F (V_x + V_{\text{wind}})^2$$

C_d : aerodynamic drag coefficient

A_F : frontal area of the vehicle: 85 ~ 95 % of vehicle width*height

ρ : air density ~1.225 kg/m³

- ✓ Frontal area includes the space under the body
- ✓ Low suspension height helps at a high speed

► Aerodynamics

- Aerodynamic drag coefficients

C _d	Model	Remark
0.137	Ford Probe V (1986)	Concept car
0.19	GM EV1 (1996)	Electric car
0.26	Opel Calibra (1989)	2.0i base model
0.27	Mercedes E230 (1996)	--
0.27	VW Passat (1997)	--
0.27	Lexus LS400 (1997)	--
0.27	BMW 318i (1998)	--

C _d	Aircraft model
0.027	Cessna 172/182
0.027	Cessna 310
0.022	Learjet 24
0.048	F-104 Starfighter
0.021	F-4 Phantom II (subsonic)
0.044	F-4 Phantom II (supersonic)
0.031	Boeing 747
0.095	X-15

C _d	Item
2.1	a smooth brick
0.9	a typical bicycle plus cyclist
0.4	rough sphere ($Re = 10^6$)
0.1	smooth sphere ($Re = 10^6$)
0.001	laminar flat plate ($Re = 10^6$)
0.005	turbulent flat plate ($Re = 10^6$)
0.295	bullet
1.0-1.3	man (upright position)
1.28	flat plate perpendicular to flow
1.0-1.1	skier
1.0-1.3	wires and cables
1.3-1.5	Empire State Building
1.8-2.0	Eiffel Tower

► Aerodynamics

- Aerodynamic drag force measurement: coast-down test
 - Vehicle slows down due to the effect of aerodynamic drag & rolling resistance
 - Neither throttle nor braking: neutral gear
 - Assume small slip i.e., longitudinal tire forces $\ll$ drag force

$$m \frac{dV_x}{dt} = -\frac{1}{2} \rho C_d A_F V_x^2 - R_x$$

- ✓ 10% higher speed $\rightarrow$ 20 % fuel loss
- ✓ 10% more weight $\rightarrow$ 5 % fuel loss
- ✓ $C_d \sim 0.3$ for typical sedans

► Aerodynamics

- Aerodynamics for racing

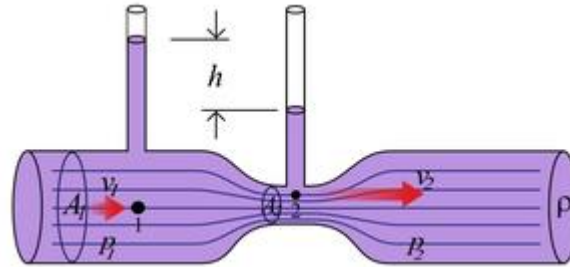
Traction force = $\mu F_z \neq \mu (m g)$

$F_z \gg m g$ due to aerodynamics

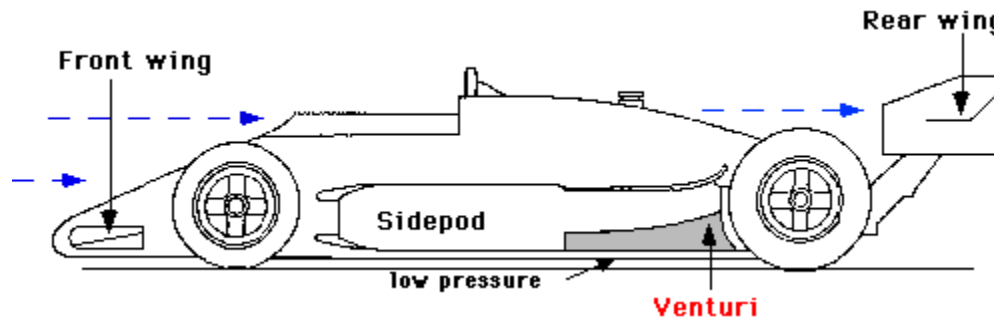


Aerodynamics

- Aerodynamics for racing: ground effect



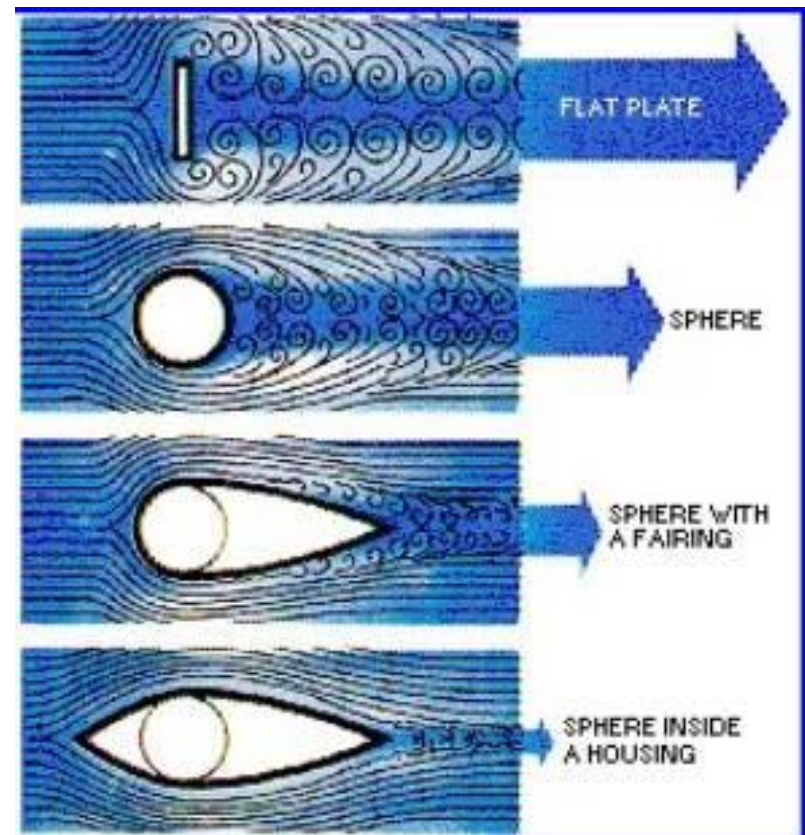
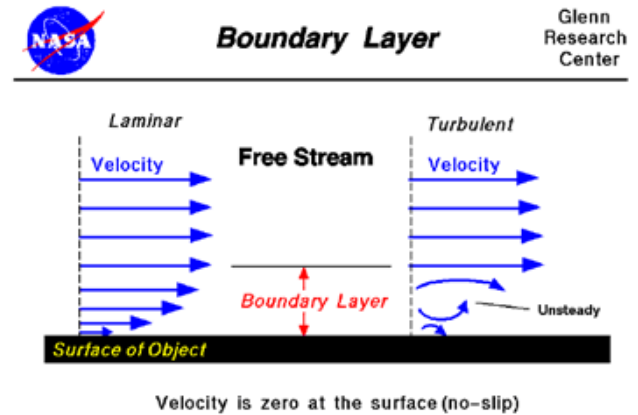
Venturi effect: one example of Bernoulli's principle



- ✓ The underbody Venturi tunnel creates a low pressure area between the chassis and the track which "sucks" the car to the track

Aerodynamics

- Causes of air drag
 - Skin friction drag: boundary layer
 - Air is viscous
 - Shear force on the boundary layer
 - Turbulent flow: Vortices mix more air into the layer
 - Form drag ~ 90%
 - High pressure air is trapped in front
 - Low pressure area behind
 - Called wake



► Aerodynamics

- Properties of air
 - Density at 1 bar (14.7 psi) & 15 °C
 - $\rho = 1.22 \text{ kg/m}^3$
 - Viscosity μ
 - Dry air = 1.80 E-5 Ns/m^2
 - Water = 1.00 E-3 Ns/m^2
 - SAE 10 oil = 1.04 E-1 Ns/m^2

$$\tau = \mu \frac{dV_x}{dy}$$

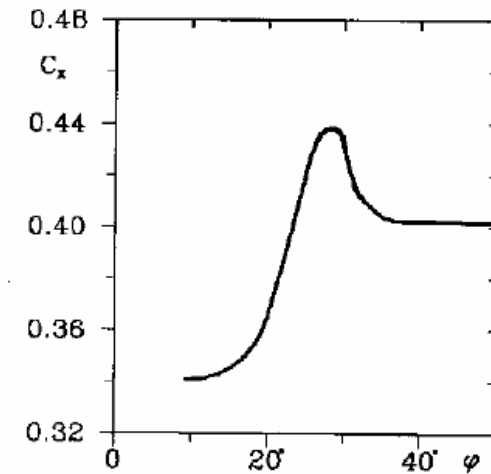
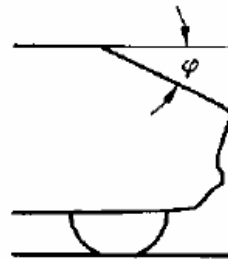
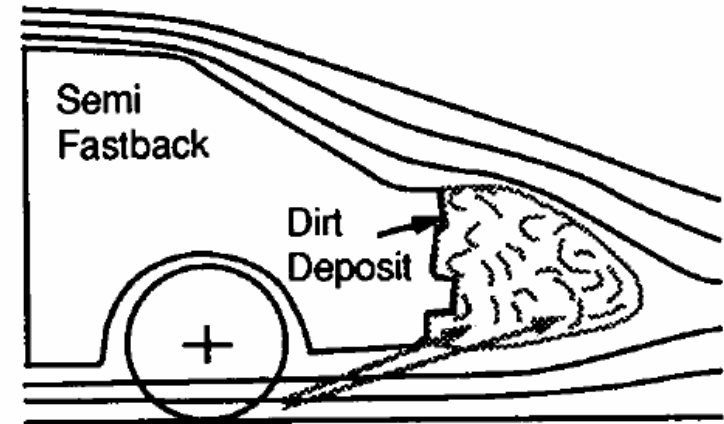
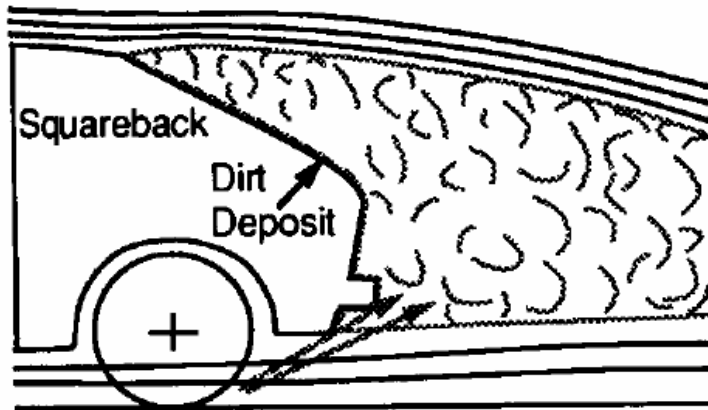
$$\tau: \text{N/m}^2$$

$$V_x: \text{m/sec}$$

$$y: \text{m}$$

► Aerodynamics

- Air drag: effect of separation point



Aerodynamics

- Reducing air drag
 - Minimize front cross section
 - Tandem seats
 - Narrow tires
 - Smaller side mirror
 - Lower suspension at a high speed
 - Ground effect
 - Aerodynamics
 - Rounded edges
 - smooth underbody
 - Narrow front & rear ends



► Aerodynamics

- Car aerodynamics: design modification example



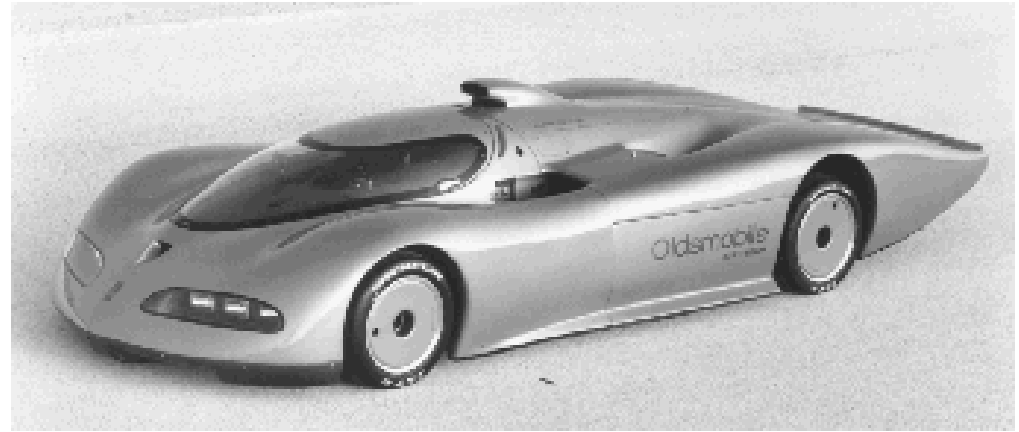
Before: 24 mpg on highway



After: 32 mpg on highway

► Aerodynamics

- Car aerodynamics: aerodynamic designs



Oldsmobile Aerotech concept car



Electrolite el-11, a 3 wheeled electrothon vehicle

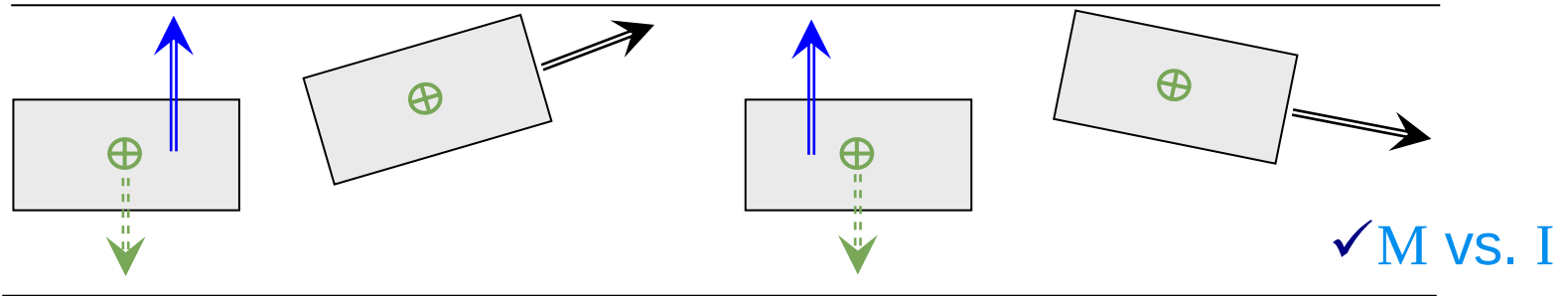
► Aerodynamics

- Air dam: A front spoiler mounted beneath the bumper and shaped to block the flow of turbulent air under the chassis
 - Increase air flow to the radiator
 - Reduce aerodynamic drag
 - Increase vehicle stability by reducing aerodynamic lift
 - Can cause more drag at a high speed



Aerodynamics

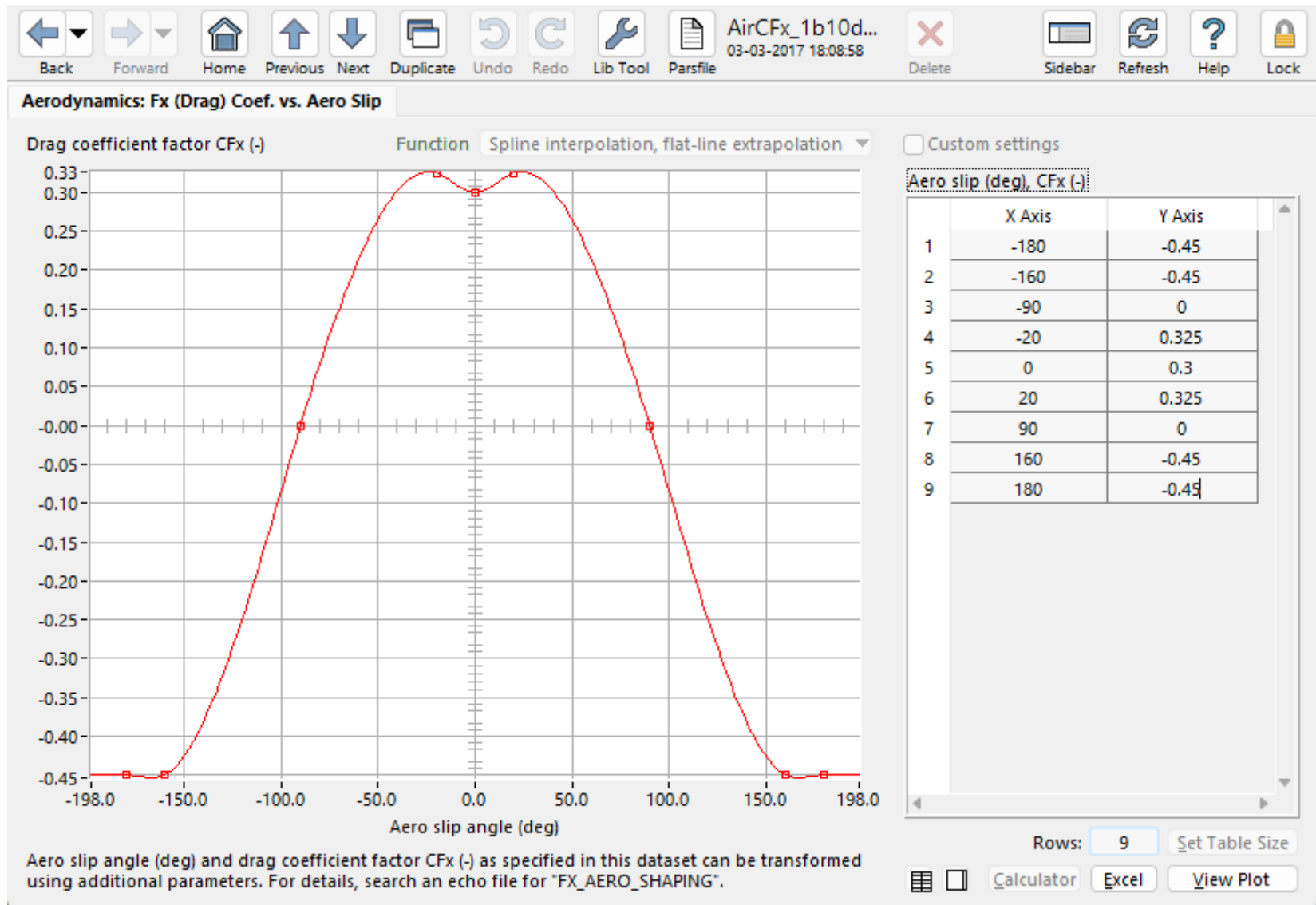
- Car aerodynamic: stability to side wind
 - Lower side drag coefficient is important too
 - Longitudinal location of pressure influence center should be located near C.G.



- ✓ Rear heavy, Low rear air influence - worst combination
- ✓ Abrupt change of the pressure influence for different angles of wind is equally bad to the stability

Aerodynamics

- CarSim Application

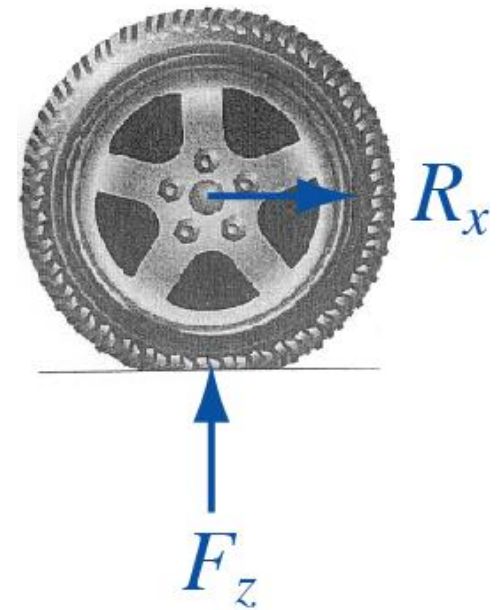


▶ Rolling Resistance (구름 저항)

- Tire rolling radius < tire radius → friction loss
- Drag is 1 ~ 4 % of the weight
- Solid wheels have no rolling resistance
- Think: Race bicycle tires
- Tire longitudinal dynamics

$$R_x = f(F_z, V_x)$$

Rolling Resistance Moment		
Rr_c	0.0038	-
Rr_v	0.000026	h/km



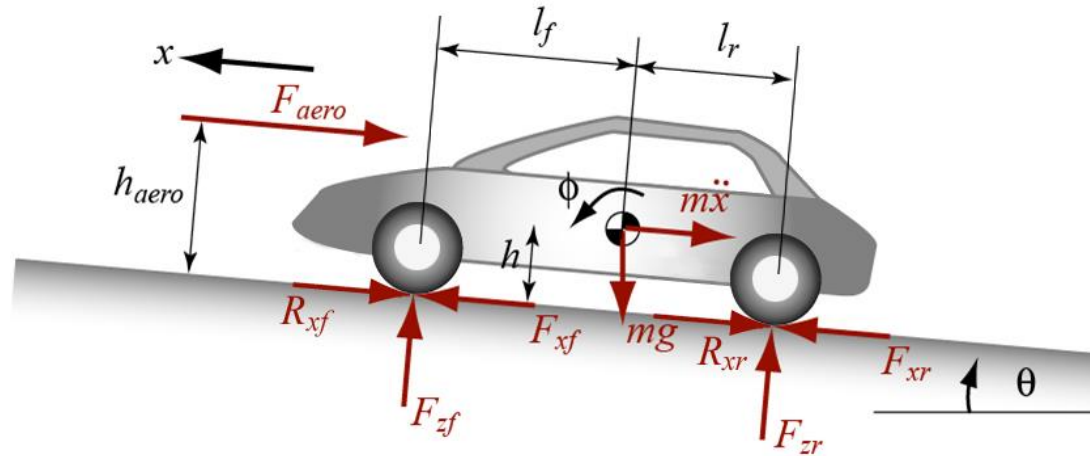
The force due to rolling resistance is implemented as a moment that opposes rotation. It is nominally described by the equation:

$$M_{yRR} = F_z \cdot R \cdot Rr_surf \cdot (Rr_c + Rr_v \cdot v_x) \quad (24)$$

where F_z is the tire vertical load, R is the effective rolling radius, Rr_surf is a parameter describing the effect of the road surface, rr_c is the constant component of the rolling resistance coefficient, and rr_v is the speed varying component.

Longitudinal Dynamics

- Equations of motion: Moment balance



$$L = l_f + l_r$$

$$I \cdot \ddot{\phi} \Big|_{\text{rear tire contact}} = -F_{zf} (l_f + l_r) + mgl_r \cos\theta - mgh \sin\theta - m\ddot{x}h - F_{aero} h_{aero} = 0$$

$$\Rightarrow F_{zf} = \frac{mgl_r \cos\theta - mgh \sin\theta - m\ddot{x}h - F_{aero} h_{aero}}{L}$$

$$F_{zr} = \frac{mgl_f \cos\theta + mgh \sin\theta + m\ddot{x}h + F_{aero} h_{aero}}{L}$$

✓ Deceleration can cause the major weight shifting

► Longitudinal Dynamics

- Weight shifting in rapid acceleration: Rear-wheel drive matters



Longitudinal Dynamics

- Drive train parts: engine, torque converter, transmission, (transfer case), drive axle, differential gear, CV joint

- ✓ No torque converter for manual transmission
- ✓ Differential gear is also for speed reduction $\sim 1:3$
- ✓ No CV joint for RWD with an unsprung solid axle

